Age and Gender Prediction using Convolutional Neural Networks

Understanding Deep Learning

age_gender_A.keras

age_gender_B.keras

words:1600

1. Introduction

This assignment focuses on predicting an individual's age and gender from facial images using convolutional neural networks. The images are taken from the UTKFace dataset, using 4,000 images for training and 1,000 for validation. Two multi-output models were developed, a custom CNN built from scratch and a pre-trained DenseNet121 model fine-tuned on ImageNet. Their performance was evaluated using Mean Absolute Error (MAE) for age prediction and accuracy for gender classification.

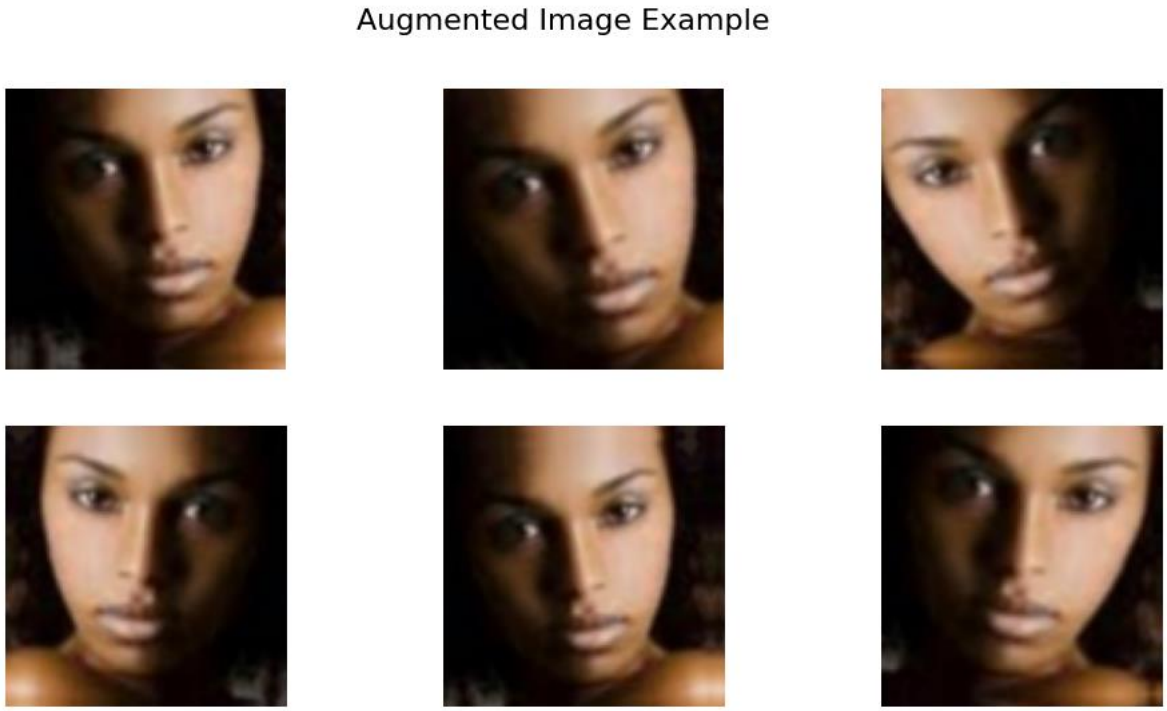
2. The custom CNN

2.1 Model Architecture

This CNN was built from scratch and its architecture follows specific guidelines, including that images are not resized (128 x 128), the size of feature maps being fed to the first fully connected layer is less than 10 x 10 (8 x 8) and the gender prediction is treated as a binary classification problem.

Data Augmentation

The first phase involves generating synthetic data from existing training examples to improve generalization through flips, rotations, zooms and translations.



Shared layers

These inputs are passed through a Conv2D layer with 16 filters of size 3x3, using ReLU activation. Batch Normalization and MaxPooling2D improve stability and reduce dimensionality. Then, there are two branches so the most important features of the facial images are prioritized for each task.

Branch 1 – Age Prediction

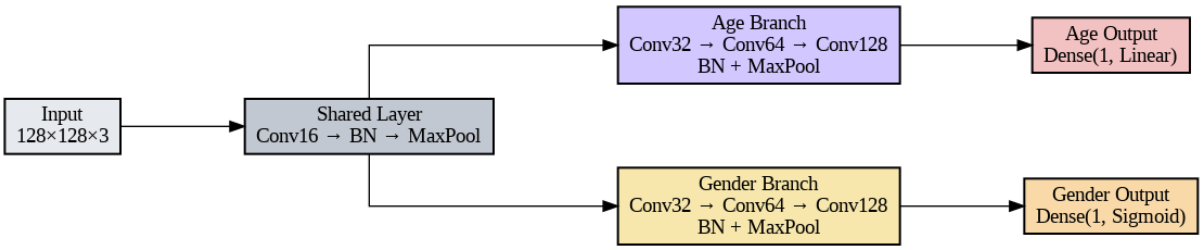
The age branch includes three convolutional blocks, each with the following structure:

- Conv2D layer (filters: 32, 64, 128)
- Batch Normalization
- MaxPooling2D

The feature maps are flattened into a single vector and passed through a fully connected Dense layer with 128 units and ReLU activation. Then, a Dropout (0.5) layer is used to prevent overfitting. The output layer is a single neuron with linear activation giving the predicted age.

Branch 2 – Gender Classification

The gender branch mirrors the architecture of the age branch, as it uses the exact same number of filters, Batch Normalization and MaxPooling layers. After flattening, a Dense layer with 128 units, using ReLU activation and Dropout (0.5) is used to pass the features. The output layer is again a single neuron with Sigmoid activation, outputting the probability of the image belonging to the "female" class, while the complementary probability corresponds to the male class.



2.2 Training Process

Optimizer

Model A was trained using the Adam optimizer with an initial learning rate of 0.0001. Compared to other optimizers such as Stohastic Gradient Descent (SGD), Adam combines momentum and adaptive gradient updates that typically lead to faster convergence.

Loss functions

Two loss functions were used: Huber (for robustness to outliers) for the age branch and binary crossentropy (for binary classification) for the gender branch, both of them treated equally (1.0 each). The training had a batch size of 32 for up to 100 epochs.

Callbacks

Two callback functions were used to improve training and avoid overfitting.

- **Early Stopping** was observing the validation loss and stopped training if there was no improvement for 10 epochs in a row and restored the weights from the best epoch.
- **Learning Rate Scheduler** reduced the learning rate by 10% after the tenth epoch.

2.3 Model Performance

Metrics

The performance of Model A is measured using the following metrics:

- **MAE** for age estimation
- **Accuracy** for gender classification

Table 1. Results of Model A after training

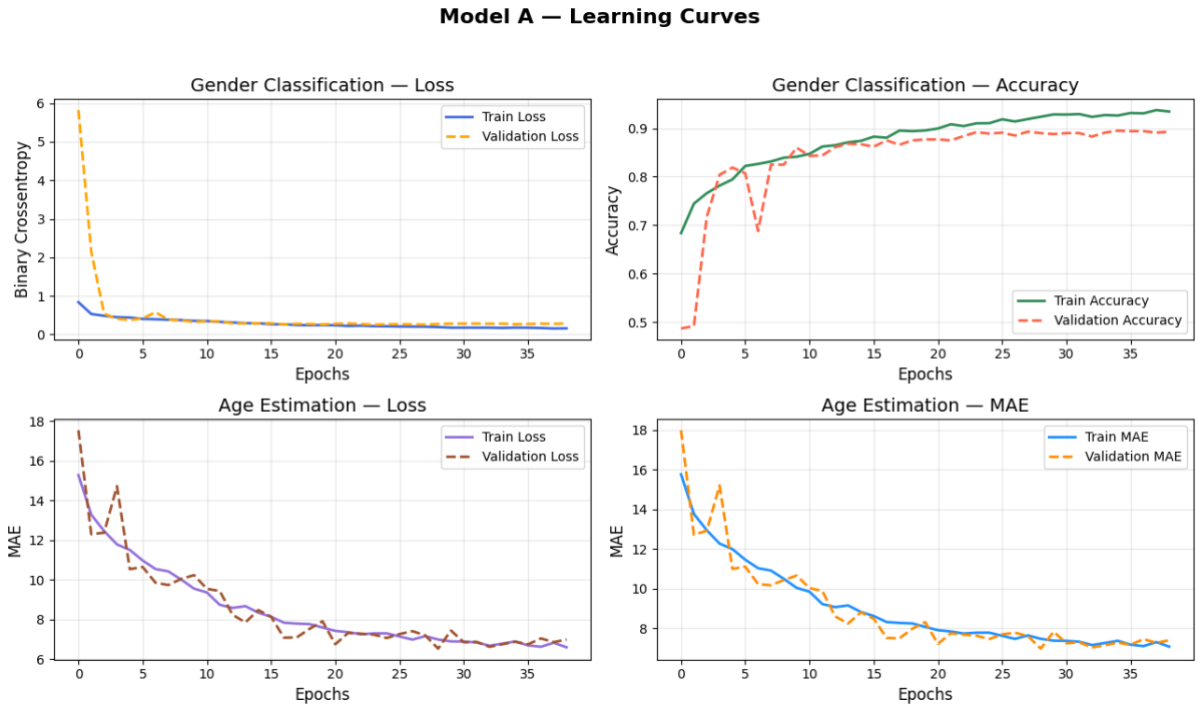
Metric	Training set	Validation set
Age MAE	5.7	6.9
Gender Accuracy	94.3%	89.0%

Age Estimation: Age MAE on the training set was 5.7, while the respective value on the validation set was 6.9. This small difference between training and validation MAE indicates slight overfitting but overall good generalization.

Gender Classification: There is a difference of 5% between the gender accuracy of the training and the validation set. This gap is normal and it seems that Model A showed mild overfitting, but it maintained high accuracy (89.0%) on unseen data achieving good generalization.

Learning curves

Figure 1. Learning curves for Model A



Age Estimation

Both training and validation Huber and MAE decrease steadily from 16 to less than 7 by epoch 35.

The curves are smooth and consistent indicating stable learning and healthy convergence.

Gender Classification

The training and validation losses drop rapidly within the first 5 epochs, while accuracy rises quickly and stabilizes around 90%.

Model A shows mild overfitting after epoch 30, but the gap between training and validation accuracy is small, indicating good generalization overall.

3. The pre-trained CNN

3.1 Model Architecture

This CNN was developed by fine-tuning a pre-trained DenseNet121 architecture from ImageNet. Similarly to Model A, images are not resized (128 x 128) and the gender prediction is a binary classification problem.

DenseNet121 is smaller than models like ResNet and EfficientNet so it is less likely to overfit within a small dataset. It learns well even with limited data, and it works even on 128×128 images.

Pre-processing

- Same **data augmentation** used in Model A.
- **BGR -> RGB** conversion is applied since images were loaded with OpenCV.
- Pixels are rescaled from **[0,1] to [-1,1]**

Backbone

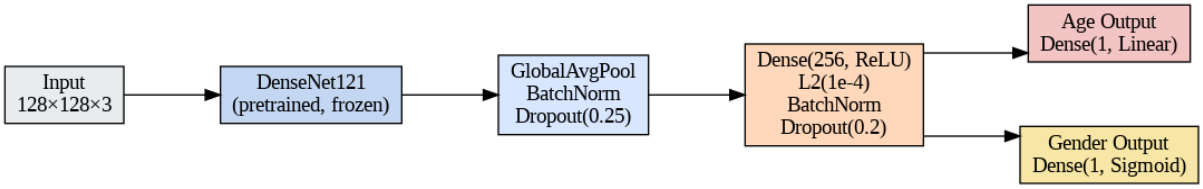
During the first training phase, the entire backbone is frozen to get the general ImageNet features. In the fine-tuning phase, the last 250 layers of DenseNet121 are unfrozen in order for the model to adapt the most important features for each task.

Custom Head

The output passes through a Global Average Pooling layer, Batch Normalization and Dropout(0.25) to reduce overfitting. Then, a Dense layer with 256 units, a ReLU activation and L2 reguralization learns high-level features and enhances generalization, followed by Batch Normalization and Dropout(0.2).

Ouput Branches

- The **age output** branch is a single neuron with a linear activation, giving the predicted age.
- The **gender output** branch is a single neuron with sigmoid activation, outputting the probability of the image belonging to the “female” class.



3.2 Training Process

Optimizer

The model was trained using the AdamW optimizer (Adam with weight decay), starting with an initial learning rate 3×10^{-4} and weight decay 1×10^{-5} during the first phase. In the fine-tuning phase, the learning rate was 5×10^{-5} and the weight decay was 1×10^{-5} in order to prevent large weight updates.

Loss funtions

MAE for the age prediction and binary crossentropy for the gender classification with same weights (1.0).

Callbacks

- **Early Stopping** watched the validation loss and stopped the training if it did not improve for several epochs restoring the best weights.
- **ReduceLROnPlateau** reduced the learning rate by half when the validation loss plateued.

3.3 Model Performance

Metrics

The performance of Model B is measured using the following metrics:

- **MAE** for age estimation
- **Accuracy** for gender classification

Table 2. Results of Model B after training

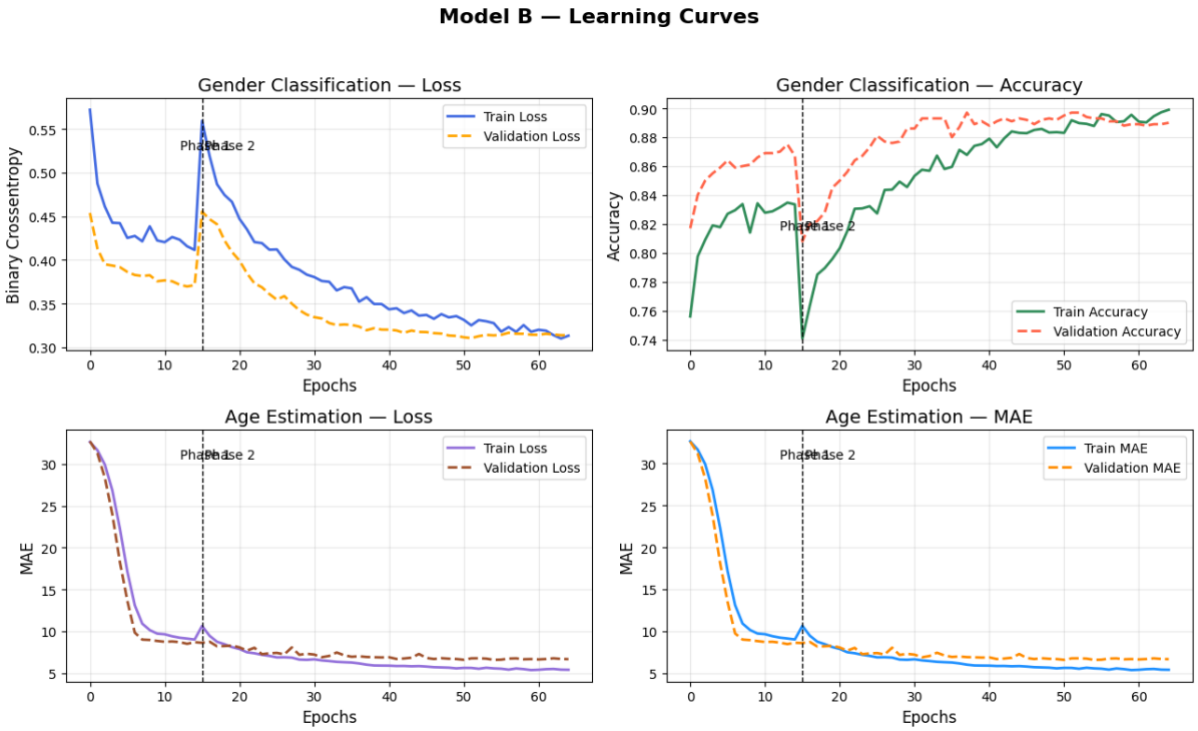
Metric	Training set	Validation set
Age MAE	3.8	6.5
Gender Accuracy	92.8%	89.3%

Age Estimation: Age MAE on the training set was 3.8, while the respective value on the validation set was 6.5. This gap indicates moderate overfitting, but reguralization kept it under control.

Gender Classification: The model achieved 92.8% accuracy on training and 89.3% on validation set. The slight difference between the two datasets indicates again strong generalization capability.

Learning curves

Figure 2. Learning curves for Model B



Age Estimation

Both training and validation losses (MAE) drop sharply during the first few epochs becoming around 5-6, indicating stable learning.

The validation curves are close to the training ones, which shows consistent generalization across epochs.

After fine-tuning begins, both losses continue to improve smoothly showing that unfreezing the backbone benefited the task.

Gender Classification

Both training and validation losses decrease consistently, with a slight bump at the transition between the phases.

Training accuracy stabilizes around 90%, whereas validation accuracy is slightly smaller (89%).

Validation loss being lower than training loss is not a sign of overfitting, but it is common due to strong reguralization.

Important note

The age estimation task shows balanced learning behavior, with both training and validation curves converging smoothly.
For gender classification, validation performance seems slightly better than training performance. This a result of data augmentation (which makes training harder), dropout (active only during training), and Batch Normalization**.** Such behavior is normal, indicating good regularization and strong generalization.

4. Summary and discussion

Both models were trained on the same dataset and evaluated using identical preprocessing and evaluation metrics. ModelB outperforms ModelA in both tasks. It achieved lower age MAE, slightly higher gender accuracy and converged faster due to transfer learning. However, ModelA showed stable learning curves and competitive performance despite being trained from scratch. Overall, ModelB generalized better but ModelA is still solid.

4.1 Strengths and limitations

Model A: Its main strengths are compactness and effective learning to the specific task that allowed it to learn important facial features even within a small dataset. Its architecture is lightweight and well suited for 128 x 128 images. However, this approach requires careful architecture design and hyperparameters selection. A limitation of this approach is that custom CNNs may not scale as effectively to larger or more diverse face datasets compared to pretrained models.

Model B: It uses pre-trained layers which already know basic image features allowing it to learn fast. Moreover, using appropriate reguralization helps the model show stable training behavior and consistent validation performance. However, its backbone is not optimized for 128 x 128 images. The model is also large making it heavy and more sensitive to overfitting on small datasets. Finally, since it was trained on general ImageNet images, it may not capture facial details as effectively, especially for more difficult tasks such as age prediction.

4.2 Potential improvements

Both models could be enhanced by additional preprocessing methods such as more varied training examples by applying stronger transformations like random cropping, brightness or blurring. These transformations can help the models become more robust to real-world variations such as lighting or facial expressions. In addition, a larger and more diverse dataset, which includes people of different ages and genders, would decrease bias and improve generalization on unseen images.

Talking about Model B, fine-tuning more layers, optimizing the learning rate schedule or experimenting with different regularization techniques could help it converge more smoothly and improve performance. Moreover, splitting the architecture earlier, like in Model A, could have helped Model B learn more specific features for age and gender prediction. Another potential improvement would be to resize the images to a larger size since DenseNet121 is trained in 224 x 224 images.

4.3 Deep Learning on real-world problems

This project highlights both the potential and challenges of applying deep learning to real-world problems. Obviously, CNNs handle image data much better than older machine learning methods as they easily learn complex visual patterns, but their success still depends on having a large and diverse dataset. Last but not least, it is important to focus on ethical considerations when applying real-world solutions, which have a significant impact, ensuring fairness and respect.

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